

3. Basics of Statistics

Where is Statistics Used in Data Science?

In the “Understanding Data” phase of data science, **statistics** is important to grasp the meaning and characteristics of the data.



Raw Data (Test Scores by Student)

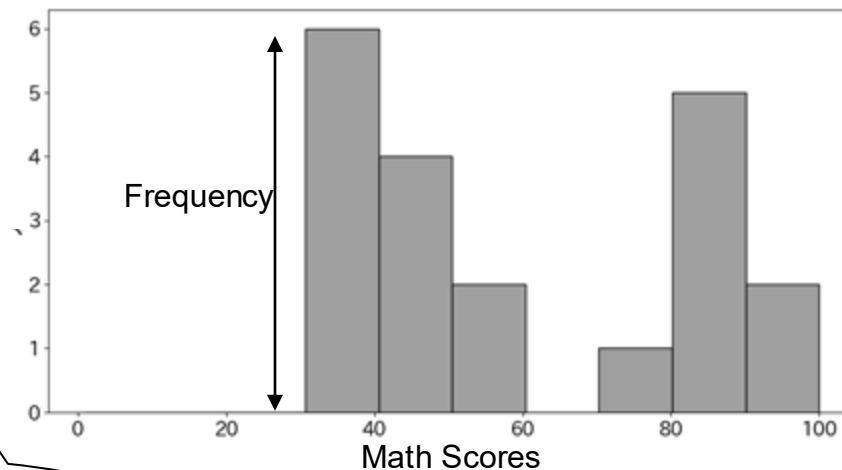
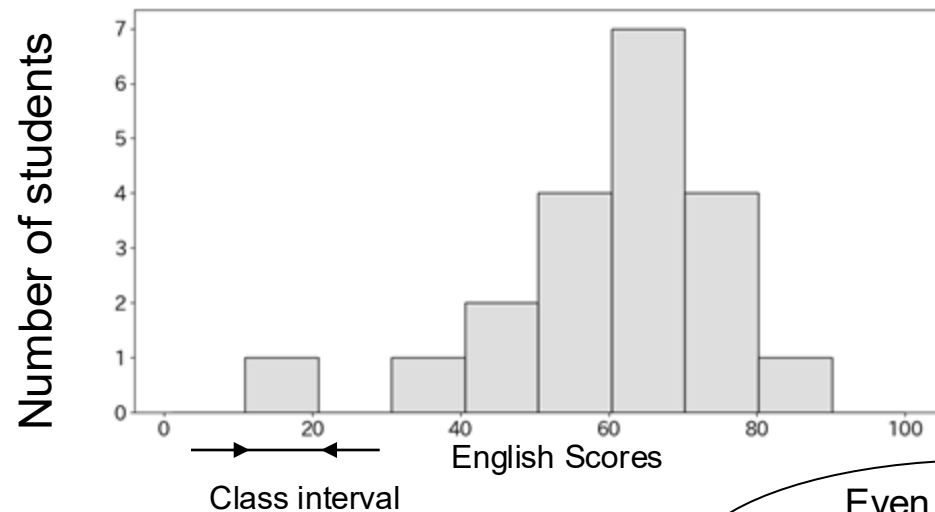
	English	Math		English	Math
Student 1	65	85	Student 11	55	90
Student 2	80	55	Student 12	55	45
Student 3	35	40	Student 13	70	85
Student 4	55	90	Student 14	50	45
Student 5	65	35	Student 15	80	40
Student 6	80	40	Student 16	70	85
Student 7	55	50	Student 17	65	50
Student 8	65	40	Student 18	80	55
Student 9	50	95	Student 19	65	100
Student 10	85	40	Student 20	15	75

Average Score in English: 62
Average Score in Math: 62

Histogram

Histogram depicts how data is distributed across various ranges. The height of the bars represent the frequency of data points in each interval.

Visualizing With a Histogram



Even with the same average score, the distribution of the scores seems to differ.

Mean & Median

Sort the test scores in ascending order and observing the **mean** (average) and **median** (middle value).

English

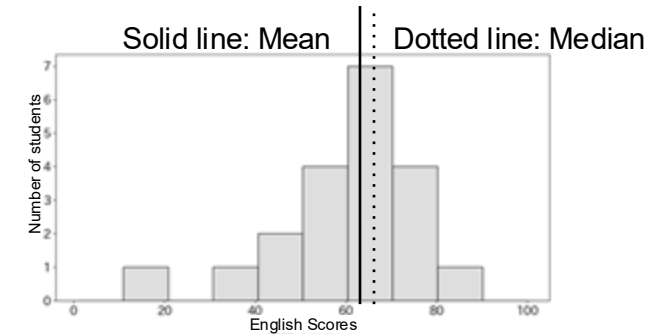
15	35	50	50	55	55	55	55	65	65	65	65	65	65	70	70	80	80	80	80	85
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

The lowest 10 scores

⋮

The highest 10 scores

Median = 65



Math

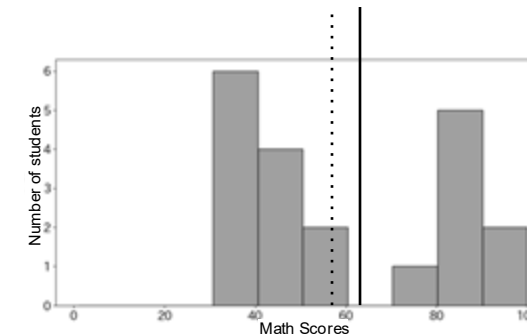
35	40	40	40	40	40	45	45	50	50	55	55	75	85	85	85	90	90	95	100
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

The lowest 10 scores

⋮

The highest 10 scores

Median = 52.5



Mean Value: Calculation Using Python

The mean is calculated by dividing the total value of the data by the number of data points. In Python, it can be calculated with Numpy.

How to Calculate the Mean

1. Add all the data points
2. Divide the total by the number of data points

$$\text{Mean} = \bar{x} = \frac{\overset{\text{Total}}{x_1 + x_2 + \cdots + x_n}}{\underset{\text{Number of data}}{n}} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example Code

```
In [1]: import numpy as np  
        np.mean([0, 2, 4, 6, 8])
```

```
Out [1]: 4.0
```

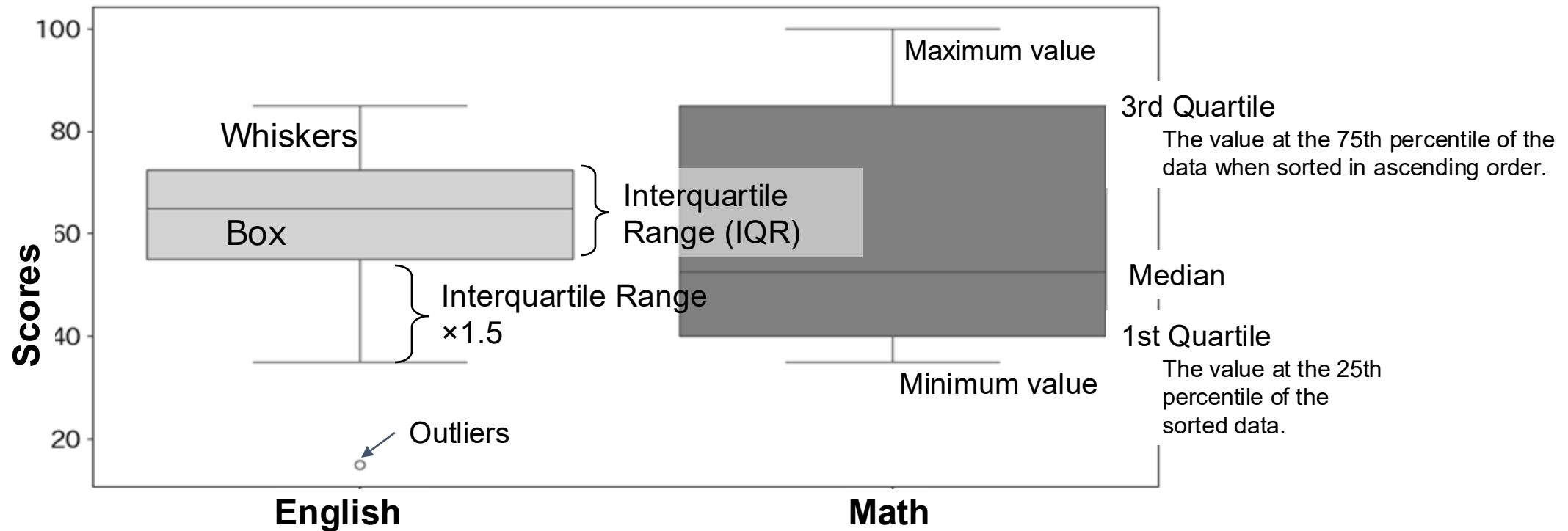
Mean

(The number obtained by adding the above values and dividing by the number of data points)

Box Plot

A **box plot** displays the data distribution, highlighting the minimum, maximum, median, and **quartiles**.

Box Plot of Test Scores by Subject



Variance

Variance is used to check the spread of data.

It helps us understand how much data deviates from the mean.

Calculating the Variance of the Test Scores

The Math test has a higher variance, meaning the scores are more spread out.

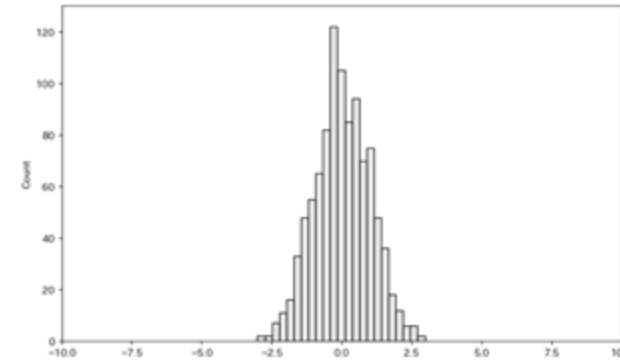
Variance of the English test : 285.3

Variance of the Math test : 524.7

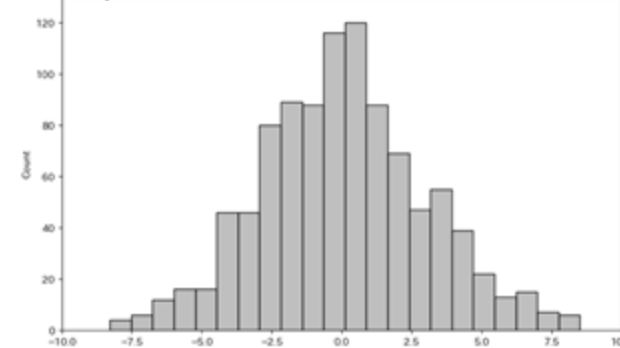
Math has more scores that are far from the average (more variance).

Concept of Variance

Small variance:



Large variance:



Variance: Calculate With Python

How to calculate variance and represent it with Python code is as follows:

Calculate Variance

1. Calculate the mean.
 2. Subtract the mean from each data point to find the deviation.
 3. Square each of the deviations to avoid negative values.
 4. Add up the squared deviations.
 5. Divide by the number of data points.
- The result is the variance (how spread out the data is)

<Formula>

$$\text{Variance} = \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

x_i : each data point, $\bar{x}$: mean, n : number of data points.

Example Code

```
In [1]: import numpy as np  
        np.mean([0, 2, 4, 6, 8])
```

```
Out [1]: 4.0    (Mean)
```

```
In [2]: np.var([0, 2, 4, 6, 8])
```

```
Out [2]: 8.0    (Variance)
```

```
In [3]: np.var([0, 2, 4, 8, 11])
```

```
Out [3]: 16.0   (Variance)
```



Standard Deviation

Standard deviation = $\sqrt{\text{variation}}$. It has the same unit as the original data.

Calculate the Standard Deviation

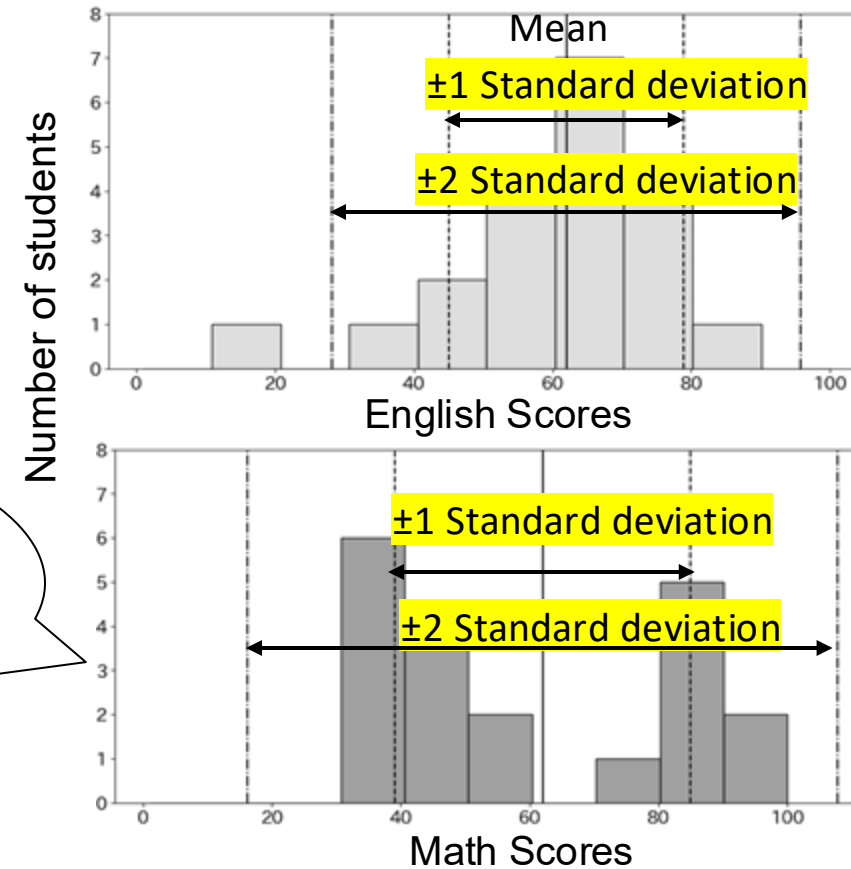
Standard Deviation of the English test: 16.9

Standard Deviation of the Math test: 22.9

Standard deviation gives a more intuitive value.

If data falls within ± 1 SD from the mean, it's considered typical. If it's beyond ± 2 SD, it's considered unusually high or low.

Standard Deviation and Data Distribution



Standard Deviation: Calculation Using Python

Calculate standard deviation and represent it with Python:

How to Calculate Standard Deviation

1. Calculate the variance
2. Take the square root of the variance

<Formula>

$$\text{Standard Deviation} = \sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

x_i : each data point

$\bar{x}$: the mean

n : number of data points

Example Code

```
In [3]: np.var([0, 2, 4, 8, 11])
```

```
Out [3]: 16.0 (Variance)
```

```
In [4]: np.std([0, 2, 4, 8, 11])
```

```
Out [4]: 4.0 (Standard Deviation)
```



Standardization

Standardization is a technique used to compare different datasets based on the same scale.

How to Standardize a Dataset

1. Subtract the mean from each data point

This clarifies how much each data point deviates from the mean

2. Divide the result by standard deviation

This clarifies how much each data point deviates from the standard deviation

Formula

$$Z = \frac{x_i - \bar{x}}{\sigma}$$

The standard value
— called the “Z-score”

x_i : each data point

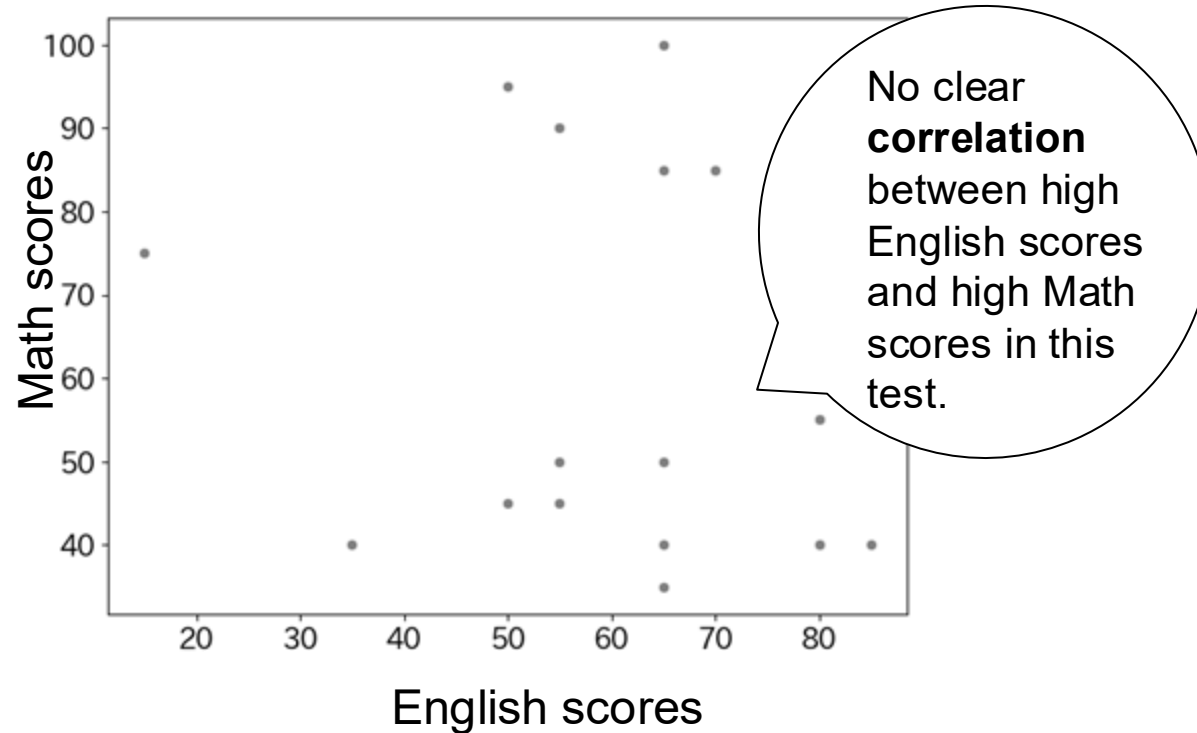
$\bar{x}$: mean of the dataset

σ : standard deviation of the dataset

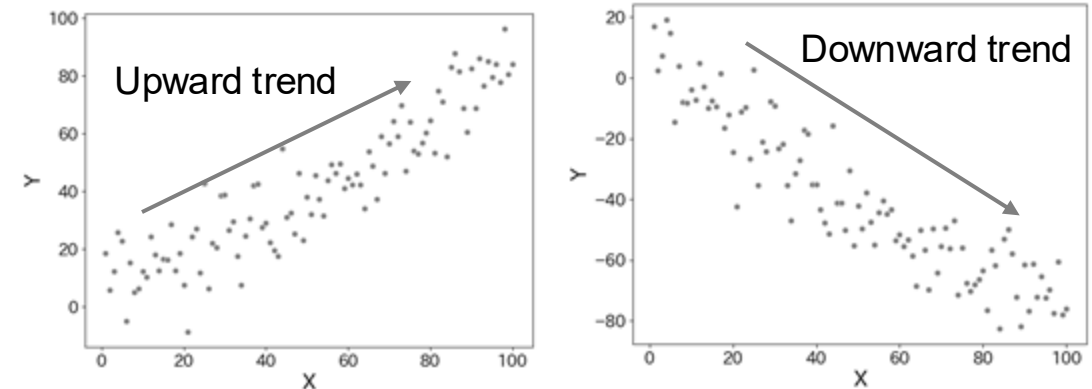
Scatter plot

A **scatter plot** depicts the relationship between two variables, helping to determine if they are **correlated**.

Scatter Plot of Our Dataset



Scatter plot of correlated datasets



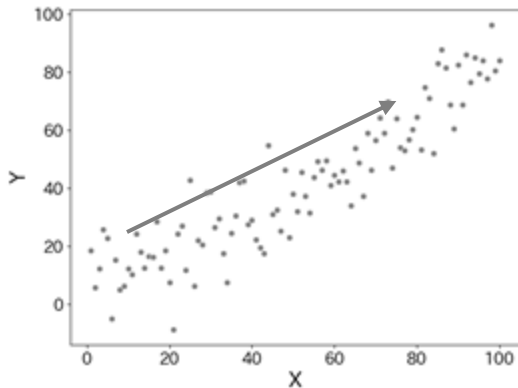
Correlation Coefficient

Correlation coefficient measures the strength of the relationship between two variables. Note: correlation does not imply causation.

What the Correlation Coefficients Tell Us

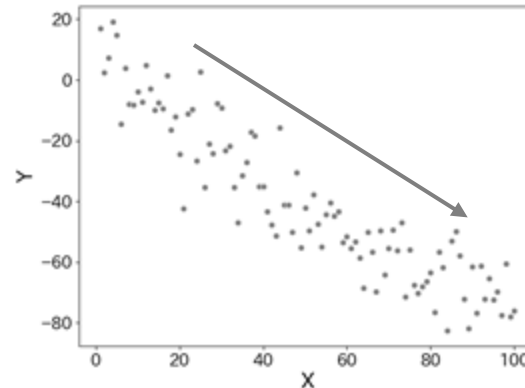
Generally > 0.7

$0 < \text{Correlation coefficient} \leq 1$
 $\Rightarrow$ "Positive correlation"



Generally < -0.7

$-1 < \text{Correlation coefficient} \leq 0$
 $\Rightarrow$ "Negative correlation"



Correlation Coefficient of Our Dataset

The correlation coefficient = -0.21

$\Rightarrow$ (almost) no correlation.

Correlation Coefficient Calculation using Python

How to calculate correlation coefficient:

How to Calculate the Correlation Coefficient

1. Calculate the mean per variable (X,Y)
2. Subtract the mean from each data point to find the deviation for X and Y
3. Multiply the corresponding deviations of X and Y
4. Sum the results of the multiplication and divide by the number of data points.
5. Calculate and multiply the standard deviations of X and Y
6. Divide 4. by the product of the standard deviations from 5.

<Formula>

$$\text{Correlation coefficient} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2}}$$

x_i, y_i : each data point

$\bar{x}, \bar{y}$: the means

n : number of data points

Example Code

```
In [1]: import numpy as np
x = [1, 2, 3, 4, 5]
y = [4, 7, 9, 10, 11]
np.corrcoef(x, y)[0, 1]
```

Out [1]: 0.9686648999069225
(Strong positive correlation)

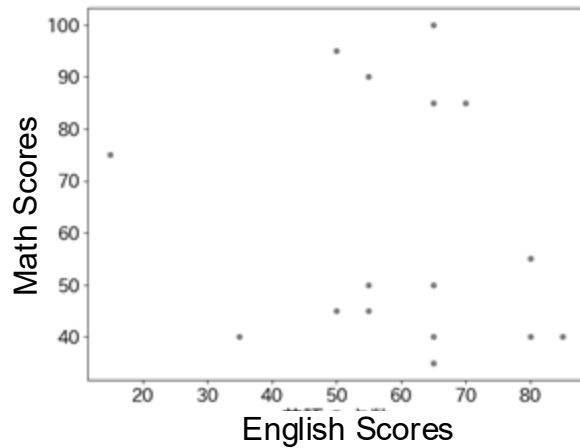
```
In [2]: x = [1, 2, 3, 4, 5]
y = [11, 10, 10, 7, 3]
np.corrcoef(x, y)[0, 1]
```

Out [2]: -0.9183996645933804
(Strong negative correlation)

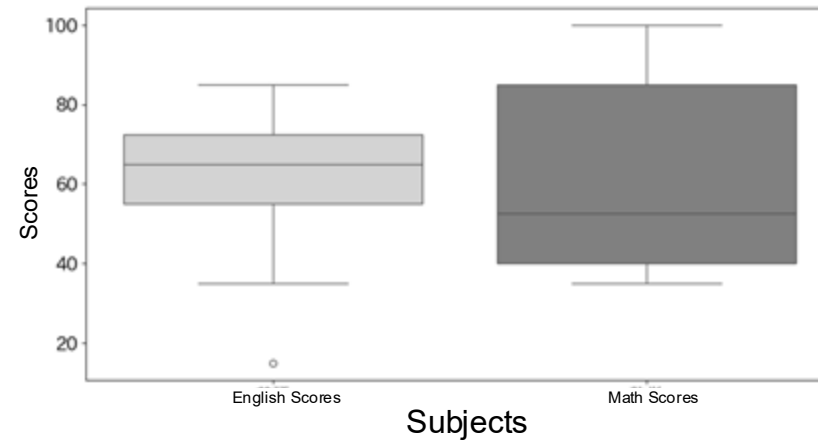
Quiz - Check Your Understanding

What are the names of the following graphs?

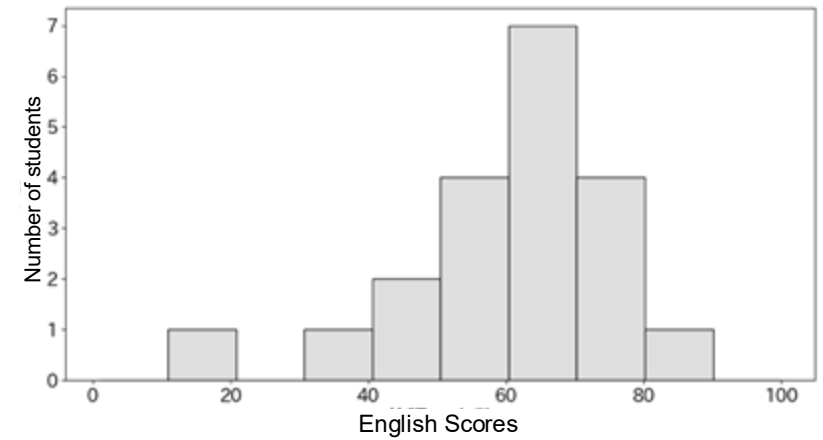
①



②



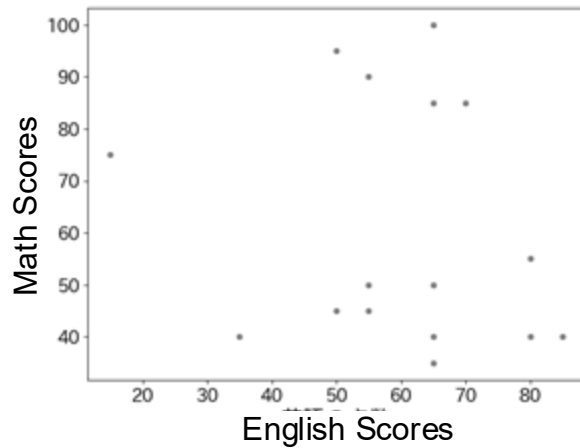
③



Quiz - Check Your Understanding

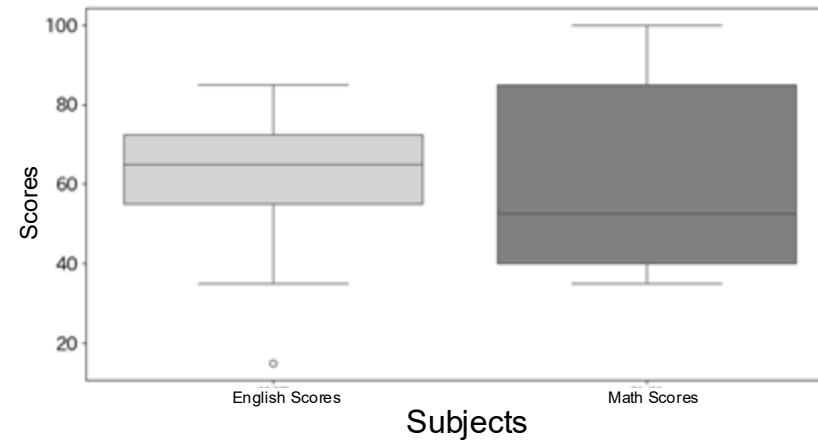
What are the names of the following graphs?

①



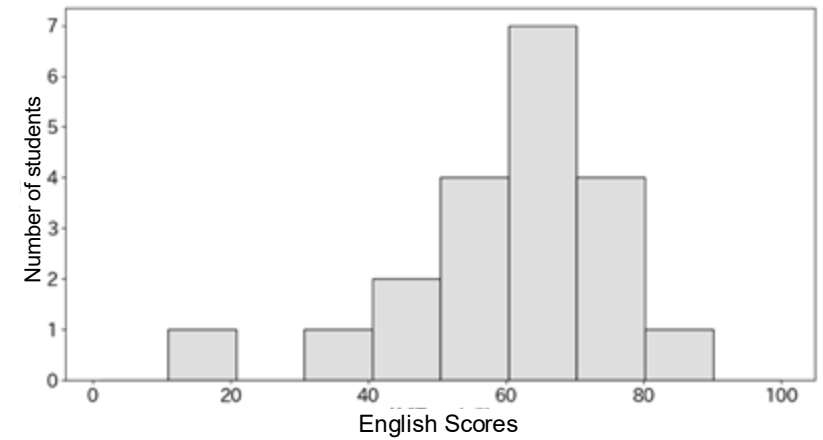
Scatter Plot

②



Box Plot

③



Histogram

Quiz - Check Your Understanding

If there are two datasets, A and B, and you want to find out the following, what should you calculate?

1. Is there a relationship between A and B?
2. Which dataset, A or B, has greater variability?
3. What is the value that lies exactly in the middle when a set of numbers is arranged in order?

Quiz - Check Your Understanding

If there are two datasets, A and B, and you want to find out the following, what should you calculate?

1. Is there relationship between A and B?

A. Correlation coefficient

2. Which dataset, A or B, has greater variability?

A. Variance or standard deviation

3. What is the value that lies exactly in the middle when a set of numbers is arranged in order?

A. Median

